Mritunjay Singh

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Education

Manipal University Jaipur **Aug 2023 – Aug 2027**
B.Tech. Computer Science and Engineering (AIML) — CGPA: 8.42 *Jaipur, India*

Blooming Dale School **May 2017 – May 2022**
Senior Secondary — 90% *Badaun, India*

Coursework

- Data Structures & Algorithms
- Operating Systems
- Network Security
- RDBMS
- Artificial Intelligence
- OOPS
- Web Development

Projects

SwasthAI: AI Diet & Recipe Assistant | *Flask, Python, JavaScript, TensorFlow, Scikit-learn* **Sep 25**

- Engineered a full-stack AI healthcare platform integrating personalized diet planning, recipe recommendation, food safety chatbot, health tracking, and food image classification.
- Designed modular Flask REST APIs for diet generation, recipe retrieval, chatbot interactions, and TensorFlow-powered image classification.
- Built a recipe recommendation engine using vectorized similarity search and nutrition datasets to provide personalized meal suggestions.
- Developed a responsive frontend with real-time health tracking, local data persistence, and accessible user interfaces for an enhanced user experience.

CodeCure AI: Explainable Drug Toxicity Prediction | *Python, Flask, React, XGBoost, LightGBM, RDKit,SHAP* **Dec 2025**

- Engineered an end-to-end AI platform for multi-endpoint drug toxicity prediction using XGBoost, LightGBM, and molecular descriptors from the Tox21 dataset.
- Design a modular Flask backend and interactive React frontend with REST APIs for toxicity prediction, explainability, and real-time visualization.
- Integrated an explainable AI pipeline using SHAP, LIME, DiCE, and RDKit structural alerts to generate interpretable predictions and comprehensive toxicity reports.

Person Re-Identification System | React, Flask, OpenCV, Deep Learning **Feb 2026**

- Built a real-time face re-identification system using YuNet CNN and SFace ResNet.
- Integrate an adaptive thresholding to reduce false positives and ambiguity.
- Integrated webcam pipelines, REST APIs, and real-time inference.
- Implemented Invisible Cloak feature using HSV color detection and morphology.

AI-Driven Phishing & Fake News Detection Suite | *Python, Streamlit, Scikit-learn, NLP, NewsAPI* **June 2026**

- Developed a unified NLP platform for phishing email detection and fake news classification using TF-IDF vectorization and machine learning models.
- Integrated NewsAPI to fetch live news articles and built an interactive Streamlit dashboard for real-time credibility analysis and prediction.
- Evaluated Logistic Regression, Random Forest, Naive Bayes, KNN, and MLP models, achieving **96.3%** phishing detection and **93.8%** fake news classification accuracy.

Experience

TechLearn Solutions **Jaipur, India**
Frontend Developer *Jan 2026 – May 2026*

- **Responsive Web Development:** Developed and maintained dynamic web applications using React.js, HTML5, CSS3, and JavaScript (ES6+), ensuring full compatibility across mobile and desktop devices.
- **Performance Optimization:** Implemented reusable UI components and optimized rendering performance using React Hooks and state management, significantly improving application speed.
- **API Integration:** Integrated REST APIs with backend services to enable real-time data updates and seamless user interactions for complex data-driven modules.
- **UI/UX Collaboration:** Collaborated with design and backend teams to enhance UI/UX consistency, improving overall application usability and user retention.

Technical Skills

Programming Languages: C++, Python, JavaScript (ES6+), SQL, C
Frameworks & Libraries: React.js, Flask, Node.js, Express.js, TensorFlow, Scikit-learn, OpenCV, XGBoost, LightGBM
Databases: MySQL, MongoDB, Prisma ORM
Web Technologies: HTML5, CSS3, Tailwind CSS, Bootstrap, REST APIs, Streamlit
Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook, Docker(basics)
Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Machine Learning, NLP, Computer Vision

Achievements & Leadership

- **Published Indian Patent** (**IN202611007521 A1**) titled *"AI-Based System and Device for Personalized Nutrition and Predictive Food Safety Assessment"*, published by the **Intellectual Property Office, India**.
- **ThinkStartup Ideathon 2025 (IIT Delhi)** – **Silver Medalist** and **Top 100 Finalist** among **16,000+ participating teams**.
- **Dean's List & Student Excellence Award (2025)** for outstanding academic performance and overall excellence at **Manipal University Jaipur**.
- **Department Student Placement Coordinator**, CSE (AIML), coordinating placement activities and facilitating student–industry interactions.
- **NCC Cadet (B Certificate)**, demonstrating leadership, discipline, teamwork, and commitment through national-level training.

Certifications

- **Oracle Academy** – Database Foundations, Database Design, SQL, and PL/SQL.
- **NPTEL** – Design and Analysis of Algorithms.
- **NPTEL** – Machine Learning.
- **Google Cloud** – Generative AI Learning Path (Fundamentals).